

Reuben Low Yu Xiang

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EDUCATION

Singapore Institute of Technology <i>Bachelor of Engineering in Robotics Systems Engineering (Honours with Distinction)</i>	Sep 2022 – May 2026
Singapore Polytechnic <i>Diploma in Mechatronics & Robotics, and Diploma Plus in Business</i>	Apr 2017 – Mar 2020

COURSEWORK

Robot Operating System (ROS1 & ROS2), Machine Learning, Reinforcement Learning, Robot Dynamics & Control, Computer Vision, Data Structures & Algorithms, Embedded Systems, System Engineering, Project Management

SKILLS

Languages: Python, C, C++
Frameworks & Tools: NVIDIA GR00T (VLA deployment), ROS1, ROS2, MoveIt2, NVIDIA Isaac Sim/Lab, Gazebo, RViz, Docker, Git/GitHub, Linux, MQTT, SolidWorks, Siemens NX, Visual Components
Hardware: Unitree H1, Unitree Go2, Kuavo V4/V4 Pro, NVIDIA Jetson AGX Orin, STM32, Arduino, Raspberry Pi, Intel RealSense, Omron TM5-700
Certifications: ROS-Industrial ROS2 Training; ROS-Industrial Consortium Asia Pacific (Sep 2025); Lufthansa-CAS Technical Training (TPE2018-1557)

WORK EXPERIENCE

Ceredroid AI Robot Learning and Motion Engineer	Jan 2025 – Dec 2025
- Developed autonomous manipulation and perception systems on the Kuavo humanoid and Unitree Go2 quadruped, spanning VLM-based task planning and evaluation, visual servoing, and vision-guided hardware payloads (Elaborated under Projects). - Collected teleoperated demonstrations on the Kuavo humanoid for NVIDIA GR00T VLA training, and deployed the trained policies on the robot. - Developed my capstone project in collaboration with the company: a Plan-Act-Correct-Verify (PACV) pipeline for long-horizon humanoid manipulation, using VLMs (Gemini Robotics ER 1.5, Cosmos-Reason1) as planners over VLAs (NVIDIA GR00T N1/N1.5). - Maintained and upgraded humanoid hardware (Kuavo, Unitree H1): replaced faulty motors, ran firmware updates and recalibration procedures, and authored maintenance SOPs adopted by the team. - Delivered interactive robotics demonstrations at SIT Open House, RoboSG, and community outreach events.	
SIT-FireFense Robotics Intern	Jun 2024 – Aug 2024
- Built an object detection pipeline for autonomous pick-and-place on an Omron TM5-700 manipulator, generating 3D pick targets from an Intel RealSense D435i depth camera. - Experimented with and integrated two detection backends into the pipeline, GPT-4o and a YOLO model. - Developed a Python driver for the Robotiq 2F-85 gripper via MODBUS protocol for the TM5-700.	
Singapore Institute of Technology Student Technical Lab Assistant	Jun 2023 – Dec 2024
- Provided technical support for robotics lab operations: soldering, wire crimping, lathe maintenance, and preparing materials and resources for robotics modules. - Performed equipment repair and maintenance on lab tools and robotic chassis used in coursework.	
Singapore Armed Forces Supply Assistant	Jul 2020 – Jul 2022
Managed unit logistical operations. Awarded Best Soldier of the Month, Sergeant Major Coin, and SBW CO Letter of Commendation.	
Advanced Remanufacturing and Technology Centre (ARTC) Engineering Intern	Mar 2019 – Aug 2019
- Optimised mirror-finish polishing on stainless steel using the ABB IRB 140: experimented with spindle speeds, tool paths, and abrasive grits; analysed and documented surface roughness and visual quality. - Integrated ABB's Simplified Robot Programming (SRP) with the ABB IRB 6660; designed 3D-printed spray nozzle adapters and demonstrated motion tracking at the 2019 ROS-I APAC conference. - Developed components for a new cable management system on the ABB IRB 6660 robot arm, reducing cable strain and preventing entanglement.	

Capstone: Hierarchical Long-Horizon Planning for Humanoid Robotics (Ceredroid AI)

Gemini Robotics ER 1.5, Cosmos-Reason1-7B, NVIDIA GR00T, Python

- Developed a long-horizon planning pipeline that decomposes natural language goals into subtasks, dispatches them to a **VLA**-driven Kuavo V4 Pro humanoid, visually verifies each step, and recovers from failures.
- Iterated from a stateless pipeline to a multi-turn conversation architecture for persistent dialogue context, and constrained VLM-generated plans to the robot's trained skill primitives to prevent hallucinated dispatch.
- Designed a comparative evaluation across 10,800 scored planning trials measuring task decomposition quality, sensitivity to scene complexity, and appropriate refusal of infeasible requests; Gemini ER 1.5 reached **97.2%** success rate vs **40.8%** for Cosmos-Reason1.
- Evaluated long-horizon monitoring on 50 **BEHAVIOR-1K** household tasks in simulation: decomposition (6.68/7), progress monitoring (2.01/3), with object hallucination the most common failure mode.

Storeman-Rosman: Autonomous Dual-Arm Shelf Picking (Ceredroid AI) | *Kuavo, ROS, YOLO, Python, REST*

- Implemented an autonomous dual-arm pick-and-place system on the Kuavo **humanoid**, with each arm independently fulfilling concurrent customer orders from a shared shelf.
- Defined function-call interfaces between the vision module and motion pipeline, dispatching trajectories from object detections through visual-servoing.
- Collected and annotated a canned-drink dataset across body and wrist camera views to fine-tune a **YOLO** detector, and enhanced the system with collision-avoidance trajectories and hard-limit safety callbacks for deployment.

Tic-Tac-Toe Humanoid Demo at RoboSG (Ceredroid AI) | *Kuavo, ROS, Computer Vision, SolidWorks, Python*

- Built an end-to-end demo in which the **Kuavo** played tic-tac-toe against the public, using a three-stage **ROS** pipeline (camera alignment, board classification, piece placement) with state management, command queuing, and blocking logic to prevent race conditions.

Autonomous Bird-Deterrence System (Ceredroid AI) | *Unitree Go2, Intel RealSense, YOLO, Arduino, SolidWorks, Python*

- Designed and built a vision-guided laser deterrence payload for the **Unitree Go2** quadruped, including the system's electrical schematic, integrating a **YOLO** model with **RealSense** depth sensing.
- Implemented **PID**-based target tracking that converts detections and depth measurements into pan-tilt angles, with multiple laser movement patterns to scare away birds.

Go2 Public Outreach Quadruped (Ceredroid AI) | *Android, MQTT, Jetson AGX Orin, SolidWorks, REST*

- Developed an Android teleoperation app for **Unitree Go2** integrating **MQTT** communication and the Unitree SportClient SDK; designed mounting hardware for the Jetson AGX Orin, speakers, and portable power to complete the deployable platform.

VLM/LLM Evaluation for Robotics (Ceredroid AI) | *NVIDIA Cosmos-Reason1, K2-Think, BitsAndBytes, Python*

- Evaluated **Cosmos-Reason1** and **Gemini Robotics ER 1.5** on synthetically generated abnormal-trajectory videos, comparing rule-based versus temporal-description prompting strategies and documenting detection failure cases.
- Quantised K2-Think to 4-bit with BitsAndBytes to fit it on a RTX 4090, measuring inference latency and VRAM use to confirm it could run locally.

EASY Pastry Printer | *STM32, C, SolidWorks, Pixy Camera, SCRUM, Project Management*

- Collaborated with **Nicher**, an artisan bakery, to develop a food printer capable of printing 2D images and logos onto a tray of pastries in any orientation autonomously.

Bolloré Logistics | *ROS1, ROS2, UR10, AgileX-LIMO, Visual Components, STM32, C, Python, SolidWorks*

- Collaborated with **3PL** company Bolloré Logistics to produce a business case for a modular automated solution that allows for easy integration with emerging technologies for Value Added Service assembly lines.
- Simulated the solution in **Visual Components** and ROS 2 (Gazebo, RViz), then built and integrated a physical **proof-of-concept** of the assembly line (robotic platform and robotic arm).
- Business case projected a **22%** increase in throughput, from **6,807** to **8,352** kit boxes per 8-hour shift, with a **25%** reduction in operational costs and headcount and a projected **ROI period of three years**.

Robotic Platforms | *C, Python, Spin Language, STM32, Pixy Camera, Parallax MCU, AgileX-LIMO*

- Developed a robotic platform capable of remote control via a GUI, further evolving it into an **autonomous system** using a Pixy camera to **follow coloured objects**.
- Implemented Breadth-First Search Algorithm in **ROS 2** Gazebo and RViz to explore an unknown maze.
- Developed scripts for autonomous goal-to-goal navigation, incorporating MoveBaseAction and MoveBaseGoal packages, implemented a GUI with **PyQt**, functionalities to save and load robot positions. Optimised navigation behaviour via ROS parameter YAML files.